How Long Is Slack Water Really? Usable Intervention Windows at the Eynhallow Sound Test Berths, 2023–2025

Isla M. Flett, Magnus Rendall and Kirsty A. Linklater

Marine Operations Research Group, Eynhallow Tidal Test Centre (ETTC), Aikerness Road, Evie, Orkney KW17 2NJ, United Kingdom

Working paper ETTC-WP-26/03 · Version 1.2 · 14 September 2026 · Supersedes version 1.1 of 2 June 2026

ABSTRACT. Every subsea task at a tidal berth has to fit inside a slack-water window. Since 2019 the Eynhallow Tidal Test Centre has planned these tasks with a single rule of thumb: a usable window of 45 minutes at every slack, centred on the predicted turn of the tide. We test it against 1,186 slack events recorded at six berths in 2023–2025 and 412 logged intervention attempts. The median usable window for a work-class ROV ranged from 19 minutes (Berth E6, spring low-water slack) to 58 minutes (Berth E1, neap high-water slack). Windows were shorter after low water than after high water and shrank further when wind blew against the flood. We propose class-specific current thresholds, a minimum predicted window per class, and a revised booking procedure in the SlackPlan scheduling tool, which the Centre adopted on 1 April 2026.

1. Introduction

Eynhallow Sound, between Rousay and the West Mainland of Orkney, reaches depth-averaged speeds of more than 3.5 m/s at spring tides, which is why the Centre placed its six grid-connected test berths there. The same currents make subsea work difficult: divers, remotely operated vehicles (ROVs) and the cable vessel can only hold position when the flow falls below a limit that depends on the type of work, and that happens only briefly around each turn of the tide.

The planning rule used since the Centre opened, a fixed 45-minute window centred on the predicted slack, was taken from a harbour-engineering manual written for sheltered estuaries. Developers have complained for several years that the rule is optimistic: of the 412 intervention attempts logged between 2023 and 2025, 97 (23.5 %) were aborted, and in 61 of those the reason recorded was current. This paper measures the usable window at each berth and tidal state and proposes a replacement for the 45-minute rule.

2. Site and data

Each berth has a bottom-mounted acoustic Doppler current profiler (ADCP) within 60 m of the device foundation. Records were averaged over 2-minute blocks and over the lowest 10 m of the water column, where divers and ROVs work; gaps longer than 10 minutes removed 74 slack events, leaving 1,186. The berths, E1 to E6, lie in a line running north-east from the Evie shore, from Berth E1 (chart depth 24 m, 0.6 km offshore) to Berth E6 (45 m, 2.1 km offshore); peak spring currents rise from 2.4 m/s at E1 to 3.8 m/s at E6.

Wind data come from the Centre's mast at Aikerness, and significant wave height (H_s) from the wave buoy moored 1.2 km west of Berth E6.

3. Defining a usable window

We define the usable window as the longest continuous period, around a predicted slack, during which the 2-minute depth-averaged speed stays below the threshold for the class of work. Thresholds, agreed with the contracted dive and ROV teams at Kirkwall in November 2025, are:

- Class D (surface-supplied diving): 0.30 m/s.
- Class O (observation-class ROV, inspection only): 0.50 m/s.
- Class W (work-class ROV, tooling and connector work): 0.75 m/s.
- Class C (cable vessel on dynamic positioning, cable lift or lay): 1.00 m/s.

Class W is the reference class for the results below because it accounts for most of the work at the site (183 of the 412 attempts). The window is measured from the moment the speed first falls below the threshold to the

Fictional document created for testing a question-answering system. All names, figures and organisations are invented.

moment it first rises above it again.

4. Results

4.1 Window length by berth and tidal state

Table 1 gives the median Class W window for each berth, separated into spring and neap tides (tidal range above or below the mean range of 2.9 m at Kirkwall) and into high-water and low-water slack. Windows shorten steadily with distance offshore, and the low-water slack is 8 to 10 minutes shorter than the high-water slack at every berth because the ebb starts abruptly. At Berths E4 to E6 no tidal state reached the old 45-minute assumption.

Table 1. Median usable window for Class W (work-class ROV, 0.75 m/s), minutes, 2023–2025

Berth	Spring HW slack	Spring LW slack	Neap HW slack	Neap LW slack	Events < 30 min
E1	47	39	58	50	6 %
E2	42	33	53	44	11 %
E3	36	27	49	40	19 %
E4	33	24	44	35	27 %
E5	29	21	41	31	38 %
E6	27	19	38	28	44 %

Windows for the other classes scale roughly with the threshold. Across all berths, the median Class D window was 0.41 times the Class W window, the Class O window 0.68 times, and the Class C window 1.37 times. A diver at Berth E6 on a spring low-water slack therefore has a median of only about 8 minutes of usable time.

4.2 Wind, swell and prediction error

Wind blowing against the direction of the flood (from the north-west sector, 280° to 340°) shortened the window most. With a mean wind of 10 m/s or more from that sector, the median Class W window fell by 31 % at Berths E4 to E6 and by 22 % at Berths E1 to E3. Wind from other sectors had no measurable effect below 14 m/s. Above H_s = 1.5 m, ROV pilots reported loss of station-keeping near the surface even when the near-bed current was within limits. SlackPlan's harmonic predictions placed the minimum current within ±6 minutes of the observed slack in 80 % of events; storm surge delayed the turn by up to 22 minutes. Teams who started at the predicted slack, rather than when the profiler showed the current had dropped, lost a median of 7 minutes.

4.3 Intervention outcomes

Table 2 summarises the 412 logged attempts by class of work.

Table 2. Intervention attempts at Eynhallow Sound, 1 January 2023 to 31 December 2025

Class	Attempts	Completed	Abort: current	Abort: weather	Abort: other	Median task (min)
D, diving	58	39	14	3	2	16
O, inspection ROV	121	104	9	5	3	14
W, work ROV	183	134	31	10	8	27
C, cable vessel	50	38	7	4	1	41
All classes	412	315	61	22	14	–

Diving had the highest abort rate (33 %), mostly at Berths E4 to E6. For Class W, 24 of the 31 current aborts occurred when the task had been planned for a low-water slack using the 45-minute rule. In those cases the

Fictional document created for testing a question-answering system. All names, figures and organisations are invented.

median planned task length was 34 minutes, longer than the observed median spring low-water window at five of the six berths.

5. Revised planning rules

From the results we derived a minimum predicted window for each class: the median task length from Table 2 plus 6 minutes for the prediction error found in Section 4.2, rounded up to the next 5 minutes. A task may be booked only if SlackPlan predicts a window at least that long for the berth, date and slack in question. Until 31 March 2026, every class used the same 45-minute assumption and could be booked at any slack; from 1 April 2026 the rules in Table 3 apply.

Table 3. Planning limits in force from 1 April 2026

Class	Threshold (m/s)	Min. window (min)	Max wind (m/s)	Max H_s (m)	Booking approved by
D, diving	0.30	25	10	1.0	Diving Superintendent
O, inspection ROV	0.50	20	14	1.5	Marine Operations Desk
W, work ROV	0.75	35	12	1.5	Marine Operations Desk
C, cable vessel	1.00	50	12	2.0	Head of Marine Operations

Two further rules apply. Diving (Class D) is not permitted at Berths E5 and E6 on any spring tide, whatever window SlackPlan predicts. When the forecast mean wind is 10 m/s or more from the 280° to 340° sector, the predicted window must be reduced by 30 % before it is compared with the minimum in Table 3.

Exception for split tasks. A Class W task too long for any single predicted window may still be booked if it is divided into parts that each fit inside one slack meeting Table 3 and the ROV can be parked on the seabed frame between slacks. The Head of Marine Operations must approve the split plan in writing at least 48 hours before the first slack. No equivalent exception exists for diving.

6. Booking procedure

This procedure replaces section 3 of the Berth Client Operations Guide (issue 7). A cancelled booking may move to another slack in the same 14-day period without repeating steps 1 to 3 if that slack also meets Table 3.

1. At least 10 working days before the planned date, the berth client enters the task in SlackPlan, giving the berth, class of work, task length in minutes and the preferred slack (HW or LW).
2. SlackPlan returns the predicted window for that class for each slack in a 14-day period; the client selects a slack whose predicted window meets the minimum in Table 3.
3. The client uploads the method statement and the dive or ROV team's competence records; the booking is then sent to the approver named in Table 3.
4. At 16:00 on the day before the work, the Marine Operations Desk checks the wind and wave forecast against Table 3, applies the 30 % wind reduction if needed, and confirms or cancels the booking by e-mail.
5. On the day, the team may put a diver or vehicle in the water only when the live reading from the berth's ADCP has been below the class threshold for 4 consecutive minutes, not at the predicted slack time.
6. The team leader reports the end of the task, or an abort and its reason, to the Desk on VHF channel 73 within 15 minutes of recovery.

References

1. Rendall, M. and Flett, I. M. (2024). *Bottom-mounted ADCP records at the Eynhallow Sound berths: processing and quality control.* ETTC Technical Note TN-24/11, Evie.
2. Eynhallow Tidal Test Centre (2023). *Berth Client Operations Guide*, issue 7. Evie, Orkney.

Fictional document created for testing a question-answering system. All names, figures and organisations are invented.